

Do **not** write solutions on this page.

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 17]

At Adam's Apple Orchard the weights of apples, W , in grams, are normally distributed with a mean 175 grams and standard deviation 8 grams.

- (a) Find the probability that a randomly chosen apple weighs less than 170 grams. [2]
- (b) It is found that 20% of the apples weigh more than w grams. Find w , correct to four significant figures. [2]

All orchards classify an apple as premium when its weight is between 170 and 185 grams.

- (c) Find the percentage of apples that are classified as premium at Adam's Apple Orchard. [2]

After orders are completed, there are many apples left over. Boxes are filled with randomly chosen left-over apples. Each box contains 40 apples.

- (d) Find the probability that a randomly chosen box contains at least 30 premium apples. [3]
- (e) If 10 of these boxes are randomly selected, find the probability that exactly 4 boxes have at least 30 premium apples. [2]

At a neighbouring orchard the weights of apples, M , in grams, are normally distributed with mean μ and standard deviation σ . It is known that:

- 82% of their apples are classified as premium
- the percentage of apples that weigh less than 170 grams is twice the percentage of apples that weigh more than 185 grams.

- (f) Find the value of μ . [6]

